

IPTIF × Inspironics Corporation

AI-Driven Hospitality Automation & Connected Facilities Architecture |
Technology Readiness Level Assessment | Pilot Proposal

Prepared for IIT-Palakkad Technology Innovation Foundation (IPTIF)



Context & Intent

Follow-up to IPTIF-Inspironics Discussion

This presentation builds on our initial conversations regarding technology collaboration opportunities between IIT-Palakkad Technology Innovation Foundation and Inspironics Corporation, focusing on practical implementation pathways.

Clarify Technology Readiness & Deployment Path

We aim to provide transparent insights into our current technology maturity levels, validation requirements, and the strategic roadmap from laboratory innovation to market-ready solutions.

Outline Pilot and Collaboration Opportunities

This document presents specific pilot project proposals, defines measurable success criteria, and explores broader partnership opportunities in semiconductor IP and embedded systems commercialization.



Problem Statement: SME/MSME Hospitality Challenges

Current State

Fragmented Guest Operations

Check-in and check-out processes remain largely manual, leading to inconsistent guest experiences and operational inefficiencies across properties.

Manual Energy & Utility Audits

Energy consumption tracking relies on periodic manual readings, making it difficult to identify waste patterns or optimization opportunities in real-time.

Underutilized IoT & BMS Data

Existing building management systems and IoT sensors generate vast amounts of data that remains siloed and underanalyzed, representing untapped intelligence.

Rising ESG Compliance Pressure

Environmental, social, and governance reporting requirements are increasing, yet SMEs lack affordable tools to measure, track, and demonstrate sustainability performance.

Opportunity



The convergence of affordable IoT hardware, cloud computing, and AI creates an unprecedented opportunity to deliver enterprise-grade intelligence to small and medium hospitality businesses. By integrating fragmented systems and applying machine learning to operational data, we can unlock 15-30% energy savings while improving guest satisfaction and regulatory compliance.



Solution Overview: Cielo Epic™



AI-Driven Platform

Machine learning algorithms continuously optimize energy usage, predict maintenance needs, and personalize guest experiences based on historical patterns and real-time conditions.



Hospitality Automation

End-to-end guest journey automation from pre-arrival room preparation through dynamic in-room controls to streamlined checkout and billing reconciliation.



Connected Facilities

Unified integration layer connecting disparate building systems—HVAC, lighting, access control, and utilities—into a single intelligent operations platform.



ESG Intelligence

Automated sustainability dashboards tracking carbon footprint, water usage, waste management, and generating audit-ready compliance reports aligned with global standards.

Cielo Epic™ is specifically designed for SME/MSME deployment, offering enterprise capabilities at accessible price points through a flexible SaaS delivery model that minimizes upfront capital investment.

System Architecture



This layered architecture ensures scalability from single properties to multi-site portfolios while maintaining data security, system reliability, and operational flexibility.

Technology Readiness Level Assessment



Core AI & Data Platform TRL 7–8

System prototype demonstrated in operational environment. Multiple successful deployments with paying customers. Continuous improvement based on production feedback.



Guest Automation TRL 7

Working prototype in relevant environment. Validated through pilot implementations across diverse property types. User acceptance testing completed with positive feedback.



IoT Integration TRL 8

Actual system completed and qualified through testing. Proven compatibility with major sensor manufacturers and communication protocols. Production-grade reliability demonstrated.



BMS Integration TRL 6–7

Technology demonstrated in relevant environment. Integration modules developed for common BMS platforms. Field validation underway with select partners to advance to TRL 8.



ESG Analytics TRL 6

Technology demonstrated in relevant environment. Reporting framework validated against international standards. Seeking pilot partners for field validation and certification alignment.

❏ **TRL Scale Reference:** TRL 6 = Technology demonstrated in relevant environment | TRL 7 = System prototype in operational environment | TRL 8 = Actual system completed and qualified | TRL 9 = Actual system proven through successful operations

Hardware Integration Model



Flexible Deployment Options

Our hardware strategy prioritizes customer choice and investment protection, accommodating diverse facility configurations and budget constraints while ensuring consistent data quality and system performance.

Leverage Existing Infrastructure

Integrate with sensors and meters already installed in buildings—energy meters, thermostats, occupancy sensors, and access control systems—minimizing additional hardware investment.

Optional Bundled Sensor Kits

Pre-configured sensor packages available for facilities lacking adequate instrumentation, ensuring comprehensive monitoring coverage without complex procurement processes.

Plug-and-Play Edge Gateways

Intelligent edge devices with automatic device discovery, local processing capabilities, and secure cloud connectivity, deployable without specialized technical expertise.

Low CapEx, High Scalability

Start with minimal upfront investment in critical areas, then expand coverage based on validated ROI, with seamless addition of new sensors and spaces over time.

Data Modeling & Visualization



Sensor-Agnostic Ingestion

Universal data collection supporting 100+ sensor types and communication protocols, with automatic format conversion and quality validation.



Normalized Asset Models

Standardized representation of physical assets—rooms, equipment, utilities—enabling cross-property analytics and benchmarking.



AI-Ready Pipelines

Automated data cleaning, feature engineering, and model training infrastructure, continuously improving prediction accuracy.



Role-Based Dashboards

Customized interfaces presenting relevant insights to each stakeholder—from technicians needing equipment alerts to executives tracking KPIs.

Operations Dashboard

- Real-time facility status
- Maintenance alerts
- Room readiness indicators
- Energy consumption by zone

Management Dashboard

- Performance trends
- Cost analysis and forecasts
- Efficiency benchmarks
- Optimization recommendations

Executive Dashboard

- Portfolio-wide KPIs
- ESG compliance status
- ROI tracking
- Strategic insights



BMS & ESG Integration



Energy Management

Comprehensive monitoring of electrical consumption across circuits, equipment, and zones. AI-driven load forecasting, demand response optimization, and automated energy reconciliation with utility billing to identify discrepancies and savings opportunities.



Waste Management

Tracking of waste generation, segregation compliance, and diversion rates. Integration with waste hauler data for complete lifecycle visibility, supporting circular economy initiatives and regulatory reporting requirements.



Automated ESG Reporting

One-click generation of sustainability reports aligned with GRI, SASB, CDP, and local regulatory frameworks. Automated data aggregation, carbon footprint calculation, and audit trail documentation for third-party verification.



Water & Resource Tracking

Real-time monitoring of water usage by source and application—domestic, irrigation, cooling towers. Leak detection through flow anomaly analysis, water quality monitoring, and conservation recommendations based on usage patterns.



HVAC Optimization

Intelligent climate control balancing comfort, energy efficiency, and equipment lifespan. Predictive maintenance alerts, occupancy-based scheduling, and integration with weather forecasts for proactive adjustments.



Compliance Monitoring

Continuous tracking against regulatory thresholds and voluntary sustainability commitments. Automated alerts for approaching limits, trend analysis for long-term planning, and documentation repository for inspections.

Guest Check-In / Check-Out Use Case

Booking @ Prediction

System receives reservation data and applies ML models to predict arrival time, service preferences, and likely room condition requirements based on historical patterns.

1

Smart Check-In

Mobile or kiosk check-in with digital key delivery, automatic room access provisioning, personalized welcome messages, and preferences loaded to in-room systems.

2

3

Automated Checkout

Energy systems scaled down, minibar and services automatically billed, digital checkout via mobile app, access credentials revoked, housekeeping immediately notified with room condition snapshot.

4

5

Room Provisioning

Automated pre-cooling/heating to guest preferences, energy systems brought online 30-60 minutes before arrival, housekeeping alerts prioritized based on predicted arrival sequence.

Dynamic Controls

Occupancy-based HVAC and lighting adjustments, energy savings during extended absences, automated service requests, integration with guest preferences for temperature and ambiance.

Quantified Benefits

- **15-25% energy savings** through occupancy-based optimization
- **40% faster check-in** reducing front desk bottlenecks
- **30% improvement in guest satisfaction** scores related to room readiness
- **20% reduction in checkout errors** and billing disputes



Pricing & Positioning

Enterprise Intelligence for SMEs

We deliver capabilities previously accessible only to large hotel chains—predictive analytics, AI optimization, integrated dashboards—at price points viable for independent properties and small groups.

Flexible Commercial Models

SaaS Subscription: Monthly per-room fees with no upfront costs

Managed Service: Full-service including hardware, installation, and ongoing support

Performance-Linked: Pricing tied to achieved energy savings, aligning incentives

Proven ROI Framework

Typical installations achieve 15-30% energy cost reduction, 10-15% operational efficiency gains, and improved guest satisfaction scores. Payback periods range from 12-24 months depending on facility size and current efficiency baseline.

Value Proposition



15-30%

Energy Savings

Typical reduction in utility costs

12-24

Months Payback

Time to full ROI

40%

Competitive Differentiation

Capability	Traditional BMS	IoT Point Solutions	Cielo Epic™
AI-Driven Optimization	Rule-based only	Limited ML	✓ Advanced ML models
Guest Experience Integration	None	Separate systems	✓ End-to-end journey
ESG Compliance	Manual reporting	Energy only	✓ Comprehensive ESG
Sensor Flexibility	Proprietary	Vendor-locked	✓ Fully agnostic
SME Affordability	High CapEx	Moderate	✓ SaaS model
Deployment Speed	6-12 months	3-6 months	✓ 2-4 weeks

AI-Native Architecture

Built from the ground up for machine learning, not retrofitted onto legacy systems

ESG-First Design

Sustainability and compliance integrated into core functionality, not afterthought add-ons

End-to-End Outcomes

Unified platform covering operations, guest experience, and sustainability vs. fragmented point solutions

90-Day Pilot Proposal

1	Objective Validate Cielo Epic™ performance in real-world SME hospitality environments, advance TRL rating from 6-7 to 8, and generate quantified case study data for broader market deployment.
2	Scope Facilities: 2-3 properties representing diverse segments (budget hotel, boutique property, small resort) Systems: Full platform deployment including guest automation, energy management, BMS integration, and ESG dashboards Coverage: 40-100 rooms per property with comprehensive sensor instrumentation
3	Duration Weeks 1-2: Site assessment, sensor installation, system configuration Weeks 3-10: Active monitoring, AI model training, baseline data collection Weeks 11-12: Data analysis, optimization verification, case study development Week 13: Final presentation and knowledge transfer
4	Key Performance Indicators Technical: System uptime >99%, data accuracy >95%, API response time <200ms Operational: Energy savings >15%, guest check-in time reduction >30% Financial: Demonstrated ROI pathway within 18 months ESG: Automated report generation meeting regulatory standards
5	Deliverables • Detailed technical performance report with sensor data validation • Quantified energy and operational efficiency gains analysis • Guest satisfaction impact assessment • ESG compliance documentation and audit trail • Scalability recommendations and deployment playbook • Academic publication-ready dataset and findings

TRL Advancement Roadmap

Current State: TRL 7

Status: System prototype demonstrated in operational environment with select commercial deployments

Evidence: Multiple successful installations, continuous production data, user feedback integration

Cluster Scale: TRL 9

Goal: Actual system proven through successful mission operations across diverse deployment scenarios

Path: Multi-property rollouts (10+ facilities), long-term reliability data (>12 months), market validation

Post-Pilot: TRL 8

Target: Actual system completed and qualified through test and demonstration in operational environment

Validation: IPTIF-supervised pilot with academic rigor, published case studies, third-party performance verification

IPTIF's Critical Role

As an academic technology innovation foundation, IPTIF brings invaluable credibility to the validation process. Your involvement provides independent verification, rigorous testing protocols, access to SME network partners, and academic publication opportunities that significantly strengthen our market position and accelerate customer adoption.

Geographic @ Market Context

Target Ecosystem



Kerala Hospitality Sector

Focus on Kerala's vibrant tourism economy with 600+ registered hotels, strong government support for technology adoption, and growing sustainability mandates.



SME/MSME Properties

Independent hotels, boutique properties, and small chains (20-100 rooms) representing 70% of Kerala's hospitality inventory but underserved by technology vendors.



IIT-Palakkad Network

Leverage IIT's regional presence, industry connections, and reputation to build trust with conservative family-owned hospitality businesses.



Market Dynamics

Risk Mitigation & Success Factors

Technical Risk: Integration Complexity

Risk: Legacy BMS systems may lack API access or use proprietary protocols

Mitigation: Sensor-agnostic architecture allows direct instrumentation bypass, pilot site selection criteria include system accessibility verification, dedicated integration team with protocol expertise

Operational Risk: User Adoption

Risk: Staff resistance to new technology, especially in family-run properties

Mitigation: Comprehensive training program in local languages, on-site support during pilot, intuitive interfaces requiring minimal training, visible quick wins (energy reports) to build buy-in

Market Risk: Price Sensitivity

Risk: SMEs may perceive solution as unaffordable despite ROI

Mitigation: Performance-linked pricing options, government subsidy alignment, IPTIF endorsement lending credibility, documented case studies with Kerala-specific cost data

Data Risk: Privacy & Security

Risk: Concerns about guest data handling and business intelligence exposure

Mitigation: Industry-standard encryption, compliance with Indian data protection regulations, transparent data governance policies, option for on-premise edge processing of sensitive data

Beyond Hospitality: Semiconductor & Automotive IP

Strategic Interest Areas



Early-Stage IP Commercialization

Inspironics seeks partnerships with research institutions developing TRL 3-4 intellectual property in embedded systems, edge computing, and sensor technologies for joint product engineering.



Embedded Systems & Edge AI

Particular interest in low-power edge AI processors, sensor fusion algorithms, and real-time data processing architectures that enhance our connected facilities platform.



Automotive Sensor Integration

Exploring applications of automotive-grade environmental sensors, LiDAR, and computer vision in smart building applications, leveraging automotive industry's reliability standards.

Collaboration Framework

1

IP Discovery & Assessment

Joint workshops to identify promising technologies within IIT research portfolio aligned with market needs

2

TRL Acceleration Program

Structured pathway from lab prototype to commercial product with shared investment and risk

3

Product Engineering Partnership

Co-development leveraging IIT research depth and Inspironics market expertise and manufacturing connections

4

Go-to-Market Strategy

Joint commercialization with revenue sharing, licensing, or spin-off entity formation based on technology characteristics

Partnership Value Proposition

Academic Validation

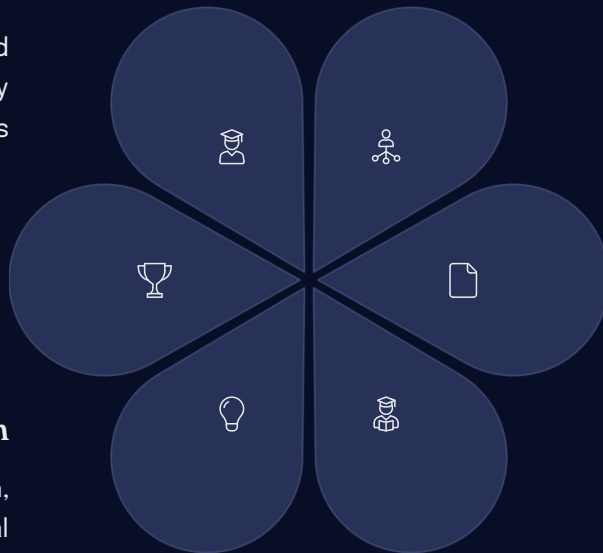
Rigorous testing protocols and independent verification lending credibility to commercial claims

Impact Demonstration

Tangible examples of research-to-market translation supporting funding and recognition objectives

Innovation Acceleration

Access to cutting-edge research, prototyping facilities, and technical expertise



Ecosystem Access

IPTIF's connections to SME networks, government programs, and industry associations

Research Publication

Joint academic papers enhancing both institutional reputation and solution credibility

Talent Pipeline

Student projects, internships, and hiring opportunities creating mutual talent development pathway

This partnership represents a model for academic-industry collaboration that advances both fundamental research and practical market impact, creating a replicable framework for technology commercialization.

Implementation Timeline & Next Steps



Bridging Academic Innovation with Real-World Impact



"The future of technology commercialization lies not in isolated breakthroughs, but in purposeful partnerships that bridge the laboratory and the marketplace. This collaboration represents precisely that vision—rigorous academic validation meeting practical deployment expertise to create solutions that serve real needs."

TRL 8

Target Achievement

Advancing technology readiness through validated pilot

90

Days to Results

Rapid validation timeline for market acceleration

15-30%

Efficiency Gains

Demonstrated energy and operational improvements

100+

SMEs Impacted

Potential reach through Kerala hospitality ecosystem